

Fine-Grained Analysis of the Hyperscan Engine

Internship Project



Supervisors: Prof. Cyril NICAUD
Dr. Pablo ROTONDO

Student: CHAU Dang Minh

September 1st, 2026

1 Motivation

2 Problem

3 Deterministic Lower Bound

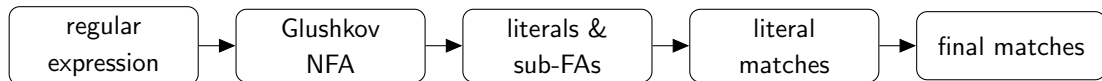
4 Stochastic Outlook

Motivation

Why Hyperscan?

- Textbook techniques analyze algorithms based on an abstraction over hardware e.g., the number of comparisons for sorting algorithms.
- Realistic analysis requires more (caching, branch prediction, SIMD ¹, etc.)
- Analyzing Hyperscan is not solely counting the number of automata transitions, since it combines
 - graph decompositions
 - SIMD literal matchers (filtering and verification)
 - SIMD automata-based regular expression verifiers
- This gives a productive source of theoretical questions.

¹Single Instruction, Multiple Data



Example: For `ab[0-9]*xy`, Hyperscan finds positions of `ab` and `xy`, then verifies the middle part with `[0-9]*`.

Guiding idea: Extracted literals are implementation details. Their lengths control how selective SIMD filtering can be.

Problem

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.

If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.

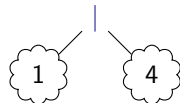
If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.



A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.

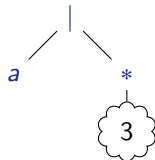


If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.

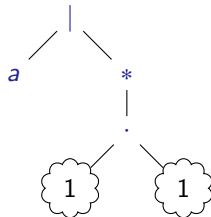


If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.

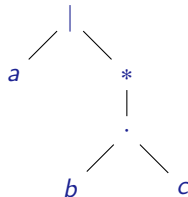


If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star (*).
- If $n \geq 3$, then the root is chosen as
 - concatenation (.) with probability u ,
 - alternation (|) with probability v ,
 - Kleene star (*) with probability $1 - u - v$.



If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

A Random Model of Regular Expressions

Let Σ be an alphabet. A BST-like distribution with parameters $u, v \in [0, 1]$ selects the root of a random tree of size n as follows:

- If $n = 1$, select a letter $a \in \Sigma$ uniformly at random.
- If $n = 2$, select a Kleene star $(*)$.
- If $n \geq 3$, then the root is chosen as
 - concatenation $(.)$ with probability u ,
 - alternation $(|)$ with probability v ,
 - Kleene star $(*)$ with probability $1 - u - v$.

If a binary operator was chosen, choose $i \in \{1, \dots, n - 1\}$ uniformly at random, then the left and right subtrees are sampled i.i.d. with sizes i and $n - 1 - i$, respectively.

Maximum-likelihood fitting on Snort rules² gives approximately $u \simeq 0.9$ and $v \simeq 0.09$.

²Cisco's Snort Rules Database <https://www.snort.org/>

The Longest-Literal Recurrence

The statistic of interest is

Y_n = length of the longest literal extracted from a random tree of size n .

Base cases:

- $Y_1 = 1$: a letter
- $Y_2 = 0$: *-rooted tree is ignored, since its language contains the empty string.

Let U_n be uniform on $\{1, \dots, n\}$. Let I and J be indicators of expectations u and v such that $I + J \leq 1$. Superscripts are used to denote independent copies.

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \underbrace{\left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right)}_{\text{take all of the subtrees}} + J^{(n)} \underbrace{\max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right)}_{\text{take the longer of the subtrees}}, \quad n \geq 1.$$

Problem: Analyze the asymptotic behavior of $\mathbb{E}[Y_n]$. In particular, is it true that $\mathbb{E}[Y_n] \sim Cn^\alpha$ for some $C, \alpha > 0$?

Deterministic Lower Bound

Bounding the Expectation

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \quad n \geq 1, \quad Y_1 = 1, Y_2 = 0.$$

Take the expectation of both sides. The difficulty is the term

$$\mathbb{E} \left[\max \left(Y_i^{(1)}, Y_{n+1-i}^{(2)} \right) \right].$$

Using

$$\mathbb{E}[\max(X, Y)] \geq \max(\mathbb{E}[X], \mathbb{E}[Y]),$$

we get a deterministic lower-bound recurrence:

$$z_{n+2} = \frac{1}{n} \sum_{i=1}^n (u(z_i + z_{n+1-i}) + v \max(z_i, z_{n+1-i})), \quad z_1 = 1, z_2 = 0.$$

Main Deterministic Result

A power-law ansatz $z_n \sim Cn^\alpha$ gives

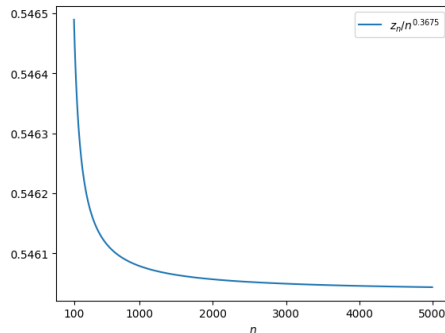
$$\alpha + \nu 2^{-\alpha} = 2(u + \nu) - 1.$$

Theorem 1 (Exact asymptotics)

If $u > 1/2$, then

$$z_n \sim Cn^\alpha$$

for some $C > 0$, where α solves the equation above.



We take the ansatz $z_n \sim Cn^\alpha$ to find out intermediate results that must be true.

Riemann-sum estimate



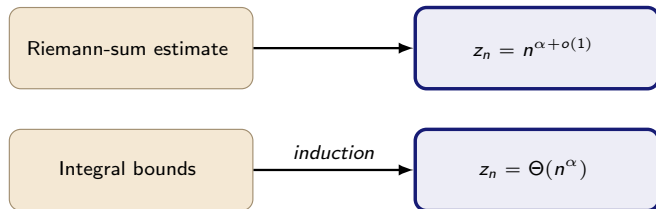
$$z_n = n^{\alpha+o(1)}$$

$$F(x, y) = u(x + y) + v \max(x, y),$$

$$\text{heuristics: } F(z_i, z_{n+1-i}) \sim F(i^\alpha, (n+1-i)^\alpha),$$

$$\frac{1}{n} \sum_{i=1}^n F(i^\alpha, (n+1-i)^\alpha) \sim n^\alpha \int_0^1 F(x^\alpha, (1-x)^\alpha) dx = n^\alpha \cdot \frac{2(u+v) - v2^{-\alpha}}{\alpha+1}.$$

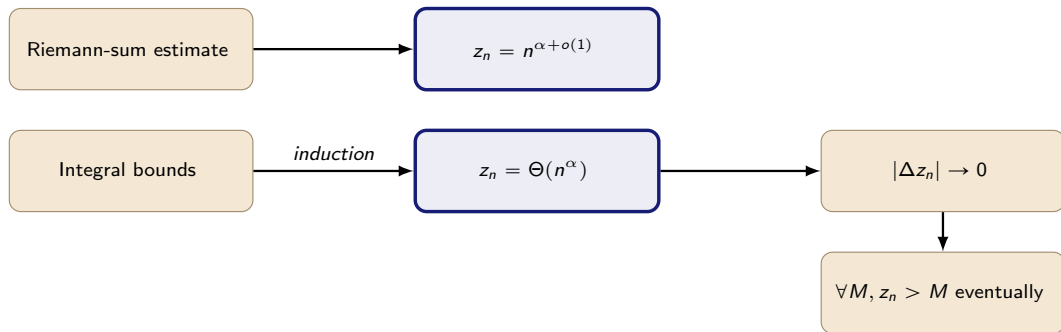
Proof Skeleton



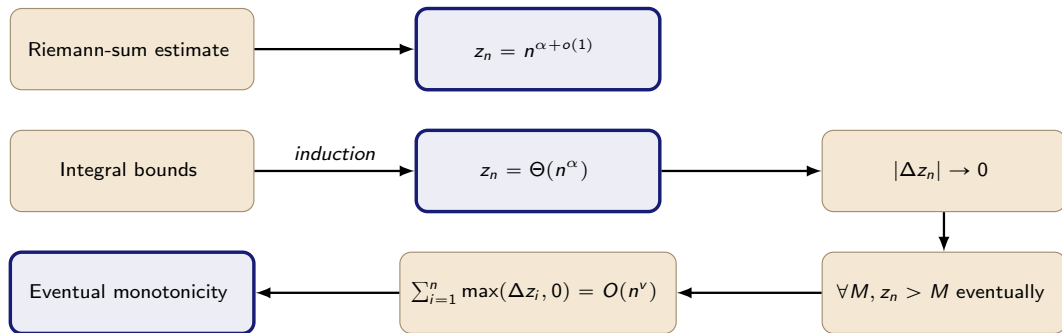
Let $f : [0, \infty) \rightarrow \mathbb{R}$ be increasing. Then

$$\sum_{i=0}^{n-1} \max(f(i), f(n-1-i)) \leq 2 \int_{n/2}^n f(x) \, dx \leq \sum_{i=1}^n \max(f(i), f(n+1-i)).$$

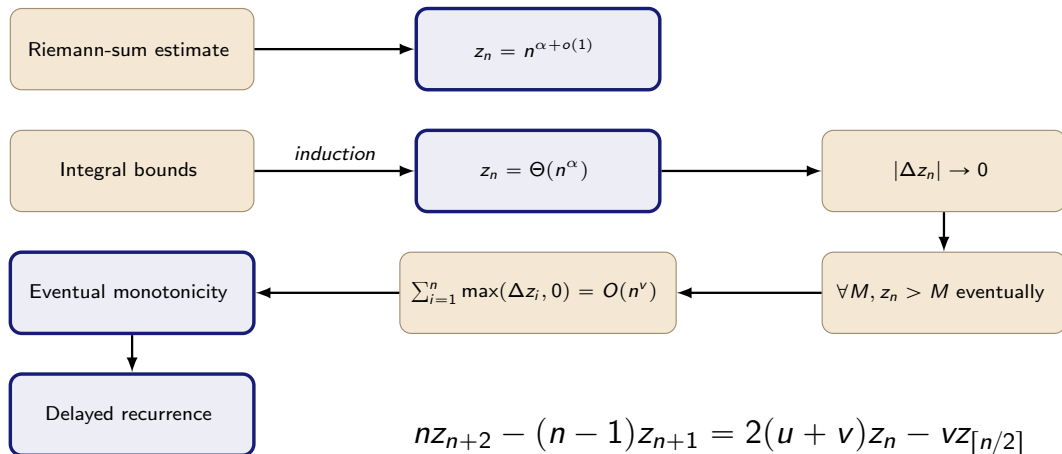
Proof Skeleton



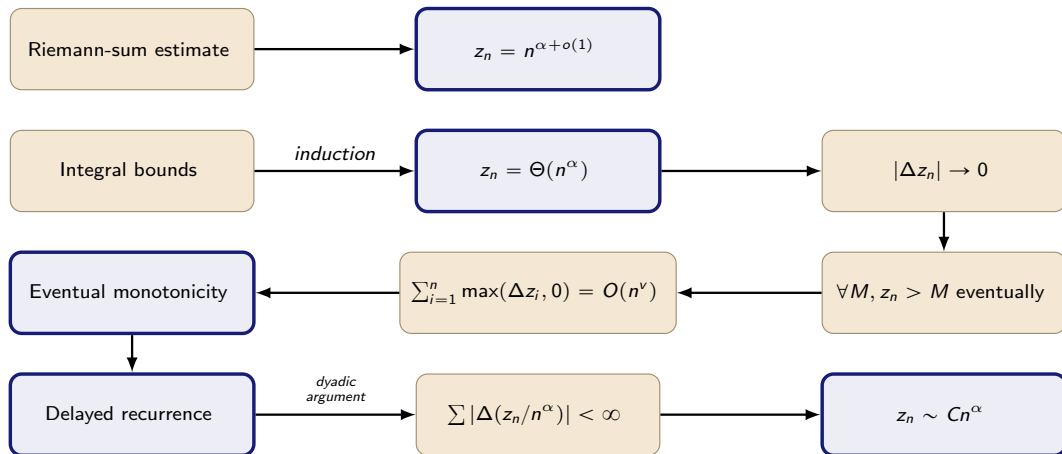
Proof Skeleton



Proof Skeleton



Proof Skeleton



Stochastic Outlook

Back to the Stochastic Recurrence

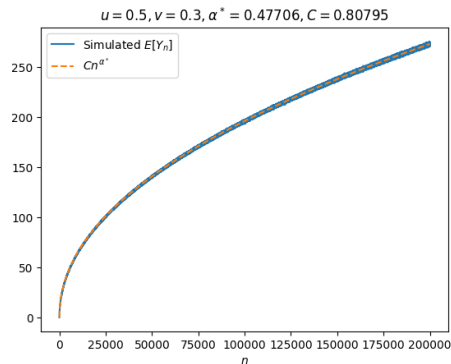
Simulations suggest that

$$y_n = \mathbb{E}[Y_n] \approx Cn^{\alpha^*}.$$

The exponent is larger than the deterministic lower-bound exponent because

$$\mathbb{E}[\max(X, Y)] > \max(\mathbb{E}[X], \mathbb{E}[Y])$$

in genuinely random cases.



A Fixed-Point Route

Normalize $X_n = Y_n/n^\alpha$. Any limiting distribution should satisfy

$$X \stackrel{d}{=} T_\alpha(X) := I \left(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)} \right) + J \max \left(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)} \right).$$

Contraction method ³:

- Consider an appropriate metric space of probability distributions.
- Show that the operator on the RHS is contractive.
- Use Banach's fixed-point theorem to get a unique solution X .
- Prove that $X_n \xrightarrow{d} X$.

Direct application failed because δ_0 is an undesired fixed point.

The workaround is a mean-preserving iteration:

$$\mu_1 = \delta_1,$$

$$\alpha_n : \mathbb{E}_{X \sim T_{\alpha_n}(\mu_n)}[X] = 1, \quad \mu_{n+1} = T_{\alpha_n}(\mu_n).$$

³Neininger, R., & Rüschendorf, L. (2005). Analysis of algorithms by the contraction method: additive and max-recursive sequences.

Theorem 2 (Tychonoff)

Let

- $\mathcal{M}$ be a locally convex topological vector space,
- $\mathcal{C}$ be a nonempty, compact, convex subset of $\mathcal{M}$,
- $f : \mathcal{C} \rightarrow \mathcal{C}$ be a continuous mapping.

Then f has a fixed point.

Theorem 2 (Tychonoff)

Let

- $\mathcal{M}$ be a locally convex topological vector space,
- $\mathcal{C}$ be a nonempty, compact, convex subset of $\mathcal{M}$,
- $f : \mathcal{C} \rightarrow \mathcal{C}$ be a continuous mapping.

Then f has a fixed point.

Preparing the ingredients:

- $\mathcal{M}(\mathbb{R})$: the space of bounded Radon measures on $\mathbb{R}$ equipped with the weak topology (our $\mathcal{M}$),
- $\mathcal{D}(1)$: probability measures on $\mathbb{R}_+$ with mean 1,
- $\mathcal{D}_2(1, M)$: elements of $\mathcal{D}(1)$ with second moment at most M (our $\mathcal{C}$),
- f is Φ , to be shown later.

Existence via Tychonoff

Mean-correcting exponent

$\forall \mu \in \mathcal{D}(1), \exists! \alpha(\mu) \in (0, 1],$

$$\mathbb{E}_{X \sim T_{\alpha(\mu)}(\mu)}[X] = 1$$

Existence via Tychonoff

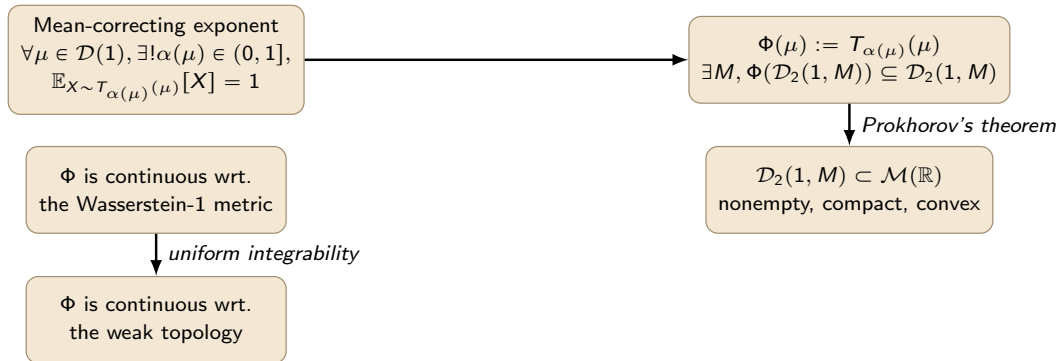
Mean-correcting exponent
 $\forall \mu \in \mathcal{D}(1), \exists! \alpha(\mu) \in (0, 1],$
 $\mathbb{E}_{X \sim T_{\alpha(\mu)}(\mu)}[X] = 1$

$\Phi(\mu) := T_{\alpha(\mu)}(\mu)$
 $\exists M, \Phi(\mathcal{D}_2(1, M)) \subseteq \mathcal{D}_2(1, M)$

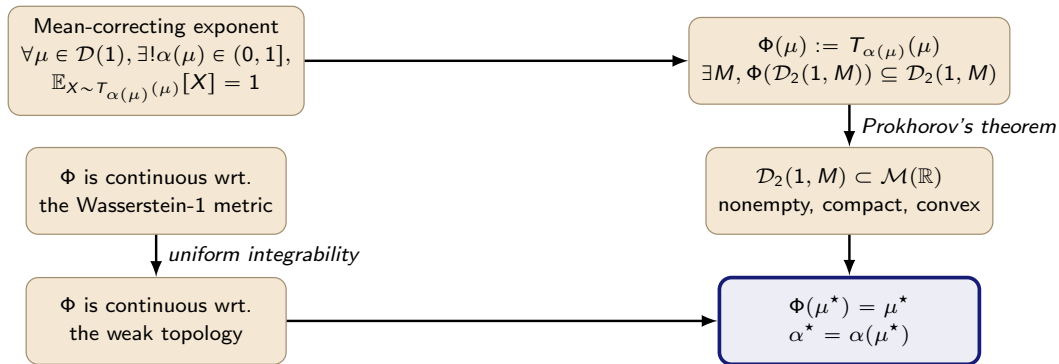
Existence via Tychonoff



Existence via Tychonoff



Existence via Tychonoff



- Hyperscan gives practical motivation for studying statistics of random regular expressions.
- The longest-literal statistic leads to a recurrence with sums, maxima, and resets.
- The deterministic lower bound has exact power-law asymptotics.
- The stochastic recurrence appears to have its own exponent α^* .

Main open direction

- Uniqueness of the fixed-point distribution μ^* .
- The asymptotics $y_n = n^{\alpha^* + o(1)}$, characterization of α^* .
- Convergence of the mean-preserving fixed-point iteration.

Thank you for your attention !